AeroStrike AI Technical Whitepaper

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Date: Oct 29th, 2026

Abstract

Accurately reconstructing a soccer ball's flight dynamics often relies on advanced tracking infrastructure, such as specialized camera installations and motion capture technology, restricting such tools to professional environments. As a result, extracting high-fidelity aerodynamic data from standard video footage remains a major limitation for standard analytical tools. In light of these challenges, this paper presents AeroStrike AI, a physics based, computer vision framework designed to model the three dimensional flight trajectory of a soccer ball from a single video. The system combines computer vision, numerical optimization and aerodynamic modelling to estimate initial launch parameters (velocity, launch angle), while accounting for aerodynamic parameters, such as drag and spin rate.

This pipeline applies computer vision to track a soccer ball's motion, extracting the location through background subtraction and contour based object filtering. Field markings and camera geometry are then used to reconstruct the detected images into real world coordinates. Initial velocity components are estimated through polynomial regression of the reconstructed coordinates, while initial spin estimates are derived from trajectory curvature. The engine then models projectile motion by accounting for gravity, aerodynamic drag and Magnus lift. Implementing the L-BFGS-B optimization algorithm refines the drag coefficient and spin rate by minimizing the error between simulated and observed trajectories.

Introduction

The ability to accurately analyze the flight trajectory of a soccer ball has grown increasingly important in modern analytics, with quantitative measurements, such as launch velocity, trajectory, and aerodynamic characteristics, providing institutions with critical data used to evaluate an athlete's performance.

Despite achieving high levels of accuracy, existing systems often rely on advanced infrastructure, including specialized hardware, motion capture equipment and optical tracking systems. Consequently, deploying these systems is cost-prohibitive and generally limited to research

laboratories. This leaves amateur athletes and youth coaches with little to no access to tools providing advanced aerodynamic analysis.

This project addresses these limitations by introducing an analytical framework that combines image based observations with mathematical modeling to recover three dimensional soccer ball trajectories. Instead of leveraging sophisticated tracking technology, the system applies fundamental geometric and physical principles to estimate the ball's motion throughout its flight.

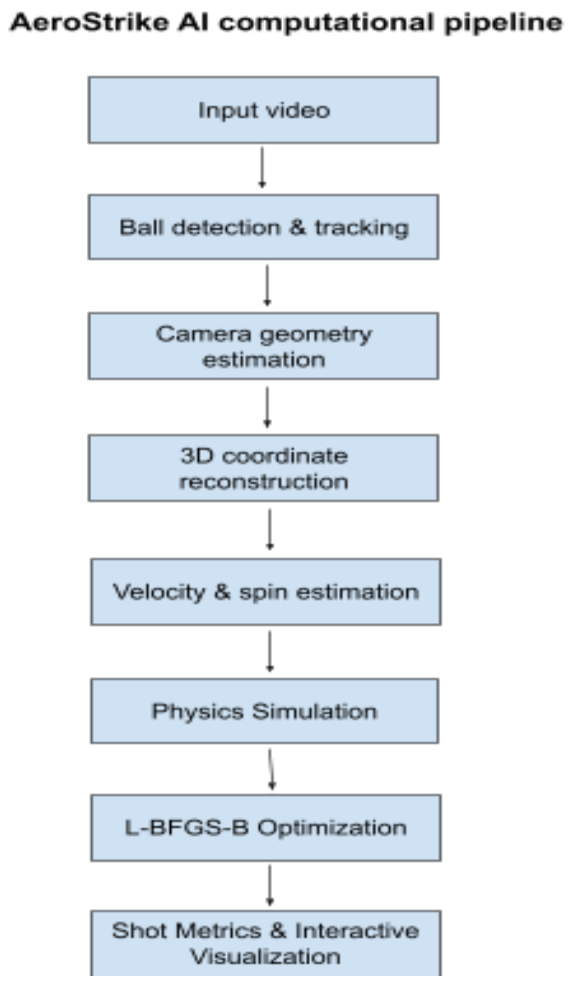
System Overview

AeroStrike AI operates as a computational pipeline that transforms monocular video recording into a physics based model resembling a soccer ball's flight. To achieve this, each task is broken down into different stages, with the outputs of one module, representing the inputs of another. This allows the framework to progressively recover the ball's kinematic characteristics.

The pipeline initially operates with the goal of identifying the soccer ball through automated ball detection, where contour analysis and geometric filtering are applied to estimate its position throughout the video. At the same time, the system extracts features of the playing field to accurately estimate camera orientation.

Image measurements are then reconstructed and projected into three dimensional coordinates using geometric transformations. Polynomial regression is applied upon these reconstructed coordinates to estimate the ball's initial components, while the trajectory curvature provides an initial estimate of the ball's spin. These estimations are critical for initializing a physics driven simulation that models the ball's motion under gravity, aerodynamic drag and Magnus lift. From here, numerical integration is implemented to extrapolate the complete flight trajectory.

Lastly, the reconstructed trajectory is compared with the simulated trajectory. A bounded L-BFGS-B optimization algorithm adjusts the drag coefficient and spin rate to minimize trajectory error between the simulated and observed flight paths. Once the optimization process is complete, the model computes quantitative performance metrics, such as launch speed, maximum height, horizontal distance and flight duration.



Computer Vision Pipeline

The computer vision module is responsible for automatically detecting and tracking the soccer ball throughout its flight. The objective in this stage is to recover the coordinates representing the soccer ball's position.

Video footage consists of several noisy elements and artifacts that, if detected, can disrupt computer vision tracking, introducing inaccuracies regarding coordinate positioning. To account for this, foreground segmentation is performed using OpenCV's Gaussian Mixture Model (MOG2) background subtraction algorithm, which separates moving objects from a static background. Morphological opening with an elliptical structuring element is applied to the resulting foreground mask to remove noise and small artifacts before implementing contour extraction.

Potential contours are extracted from the processed foreground mask and narrowed down based on geometric criteria. Contours below a minimum area threshold are discarded to eliminate insignificant detections, while the remaining contours are evaluated using their bounding-box aspect ratio to favor approximately circular objects. Only contours satisfying all geometric constraints are retained as valid candidates.

For each detection, the centroid and ball diameter are estimated. When sufficient contour points are available, an ellipse is applied to accurately estimate the ball’s diameter, otherwise, a minimum enclosing circle is used as a fallback. The detected ball diameter is then used to determine the image scale for later three dimensional reconstruction.

The resulting measurements from this stage provide the geometric observations needed for the trajectory estimation stages.

Camera Geometry and Three Dimensional Reconstruction

Before performing aerodynamic analysis, the tracked pixel coordinates must be transformed into meaningful three dimensional world coordinates. Furthermore, with input consisting of a single monocular video, depth information is not directly available and must be inferred through geometric reasoning.

To determine the camera’s spatial orientation, features of the soccer field are extracted using edge detection, along with the probabilistic Hough Line Transform. The detected line segments are filtered based on their geometric properties, with the goal of isolating prominent field markings. Following this, pairwise interactions are computed to estimate the horizontal vanishing point, which provides information regarding the camera's viewing direction. The estimated vanishing point is also used to approximate the camera's yaw and pitch angles, relative to the field. These parameters are incorporated into a three dimensional rotation matrix, which allows coordinates within the camera's reference frame to be transformed into a world coordinate system.

Depth estimation is performed using a perspective projection model. When a reliable estimate of a ball’s diameter becomes available, depth is computed from the known diameter of a standard soccer ball, with the camera's focal length being derived from the horizontal field of view. If the ball's size cannot be estimated accurately, the framework approximates depth using the soccer ball's position relative to the horizon, defined by the estimated vanishing point.

The reconstructed depth and camera orientation are then combined to transform the detected image coordinates into their corresponding three dimensional world coordinates, using a pinhole

camera model. These coordinates serve as the geometric representation of the observed flight path.

Physics Flight Model

After estimating the ball's initial position and velocity, its flight is modeled through the use of fundamental physics principles. This project treats the soccer ball as a rigid body, influenced by three primary forces: gravity, aerodynamic drag and Magnus lift. These forces determine the ball's acceleration throughout its flight.

The motion of a ball can be described by Newton's Second Law (F = MA), which states that the net force acting on an object is equal to the product of its mass and acceleration. This proposed framework calculates net force as the sum of the gravitational, aerodynamic and Magnus forces.

Gravity acts as a constant downward force, with an acceleration of 9.81 m/s². Aerodynamic drag opposes the direction of motion, while increasing proportionally to the square of the ball's speed. The magnitude of the drag force is dependent upon several factors, including air density, the drag coefficient and the cross-sectional area of the ball.

This project employs a Magnus force model to capture the effects of the ball's spin. The Magnus force acts perpendicular to both the ball's velocity and its axis of rotation, producing lateral curvature, commonly observed in free kicks.

The combined effects of gravity, aerodynamic drag and Magnus lift constantly change the ball's velocity throughout its flight, with all of these forces varying significantly as the ball moves. To account for this, the simulation uses SciPy's Runge–Kutta (RK45) numerical integration method to calculate the ball's position and velocity over many small time intervals. This process is repeated until the flight time has elapsed or the ball reaches the ground, producing a three dimensional trajectory with the acting forces.

Governing equations:

$F_{net} = ma$

$F_g = mg$

$F_d = -\frac{1}{2}\rho C_D A|v|v$

$F_m = \frac{1}{2}\rho A C_L |v|^2 \hat{n}$ → where $\hat{n}$ is the unit vector perpendicular to both the ball's velocity and its axis of rotation, defining the direction of the Magnus force.

Aerodynamic Parameter Optimization

Despite providing estimates of the ball's three dimensional motion, computer vision fails to consider the aerodynamic properties acting upon the flight. For instance, the drag coefficient and spin rate cannot be measured from the input video and must instead be estimated computationally.

Optimization starts with the initial velocity components being estimated from the reconstructed trajectory using polynomial regression. Polynomial functions are assigned to the ball's horizontal, vertical and depth coordinates, as functions of time. This results in the formation of a fitted curve, which provides estimates of the initial velocity in all spatial directions.

Analyzing the curvature of the reconstructed trajectory provides an initial estimate regarding the ball's spin, where changes in the ball's lateral motion are used to approximate the influence of the Magnus effect. This establishes a physically realistic baseline for subsequent optimization. These estimated parameters are supplied to the physics flight model, which generates a simulated trajectory. This simulated trajectory is then compared to the observed flight path.

To improve the accuracy of the simulated flight path, AeroStrike AI implements the Limited-memory Broyden–Fletcher–Goldfarb–Shanno with Bounds (L-BFGS-B) optimization algorithm. The optimizer iteratively adjusts the drag coefficient and spin rate, while minimizing the Mean Squared Error (MSE) between the simulated and reconstructed trajectories. Throughout this process, parameter bounds are applied to ensure physically realistic estimates are produced. Upon finishing, the optimized aerodynamic parameters are used to generate the final flight simulation.

Optimization objective:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N} ||Pi^{obs} - Pi^{sim}||^2$$

Performance Metrics

To summarize the characteristics of a soccer strike, the framework introduces several quantitative metrics. These metrics provide a detailed description of the shot, while enabling comparisons between different trajectories.

The framework considers the following metrics:

- Launch speed
- Launch angle
- Maximum height
- Horizontal distance
- Flight time

These quantities are directly calculated from the optimized trajectory and initial launch conditions, providing a concise numerical summary of the shot.

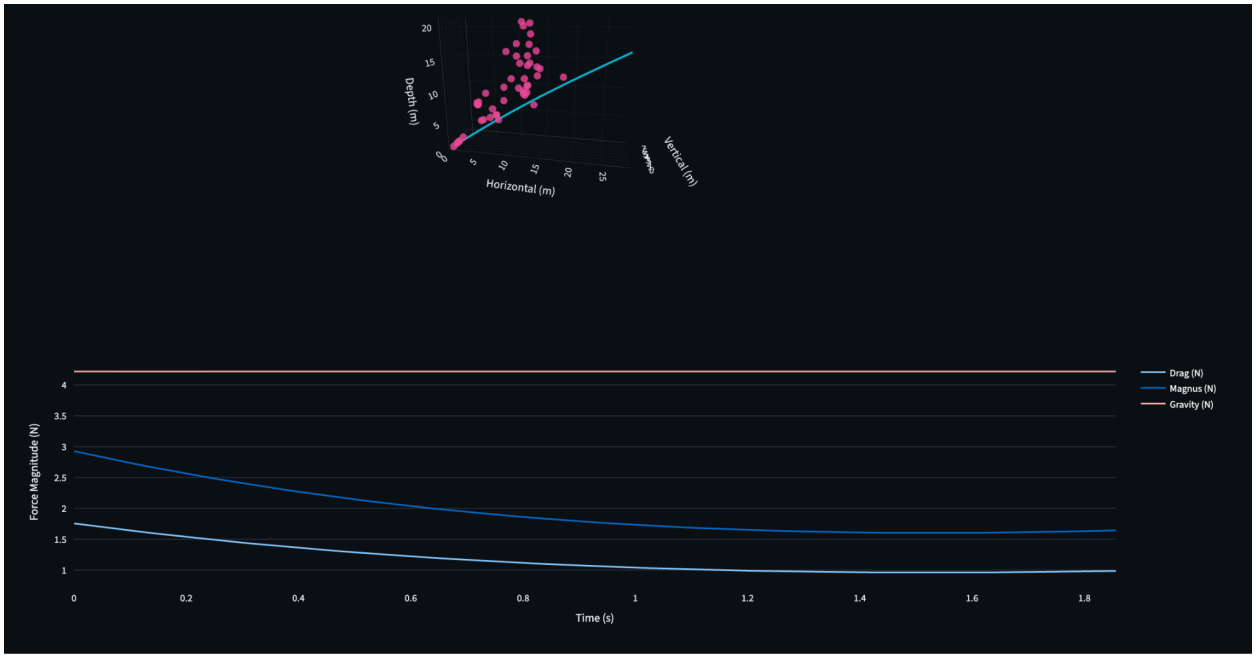
Results

To evaluate the performance of AeroStrike AI, the framework was tested on various soccer strike videos captured from a single monocular viewpoint. This paper analyzes two of these videos, focusing on a low driven, 70mph shot, and a Cristiano Ronaldo free kick. The analysis compares the launch speeds between the predictive model and reference video, while detailing other key estimations such as launch angle, peak height, horizontal distance and flight time.

Test case 1 - Controlled 70 mph low driven shot:

Metrics	**Reference video**	**AeroStrike AI Prediction**	**Absolute Error**	**Accuracy**
Launch Speed (m/s)	31.3 m/s	22.5 m/s	8.8 m/s	71.9%
Launch Angle (°)	N/A	21.7°	N/A	N/A
Peak Height (m)	N/A	4.34 m	N/A	N/A
Horizontal Distance (m)	N/A	32.84 m	N/A	N/A
Flight Time (s)	N/A	1.86 s	N/A	N/A

***N/A indicates data is not available for that metric**

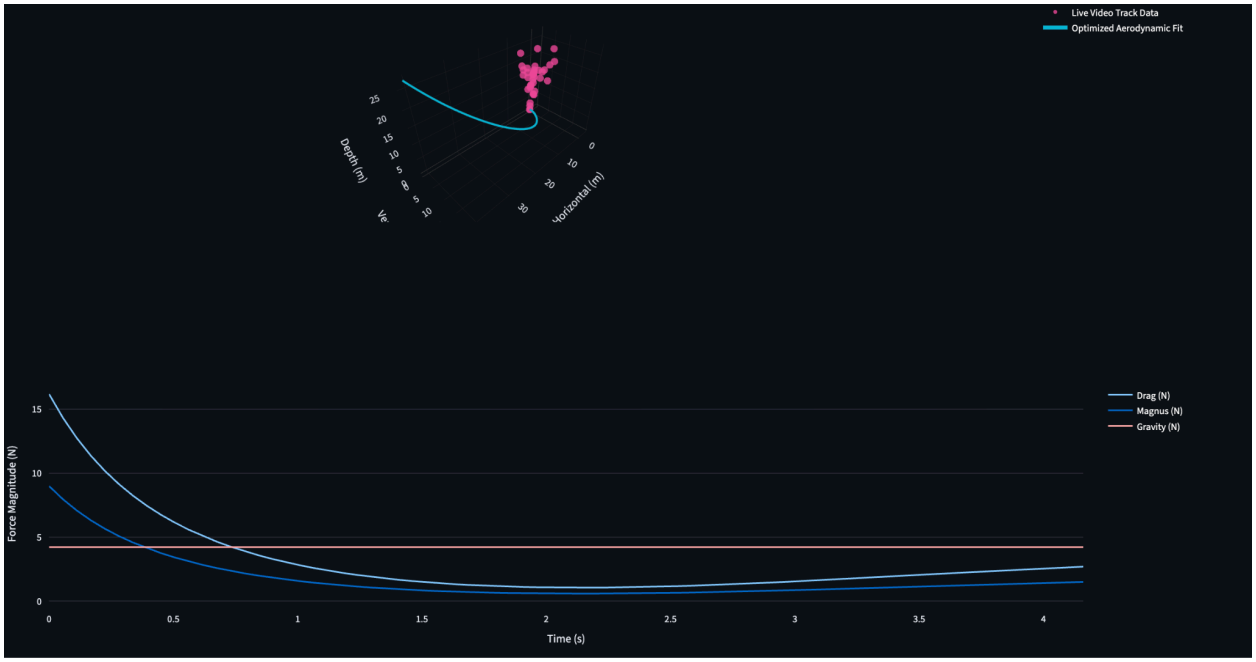


This first test case depicts inaccuracies in launch velocities, with the reference launch speed being 31.3 m/s and the model estimating a velocity of 22.5 m/s. This produced an absolute error of 8.8 m/s and an accuracy of 71.9%. Despite deviation in the velocity estimate, the reconstructed trajectory remained physically consistent with the observed shot characteristics.

Test case 2 - Cristiano Ronaldo free kick:

Metrics	**Reference video**	**AeroStrike AI Prediction**	**Absolute Error**	**Accuracy**
Launch Speed (m/s)	36.1 m/s	39.3 m/s	3.2 m/s	91.14%
Launch Angle (°)	N/A	39.2°	N/A	N/A
Peak Height (m)	N/A	19.62 m	N/A	N/A
Horizontal Distance (m)	N/A	48.75 m	N/A	N/A
Flight Time (s)	N/A	4.16 s	N/A	N/A

***N/A indicates data is not available for that metric**



The second test case demonstrates high accuracy in launch velocities, with the reference launch speed being 36.1 m/s and the model estimating a velocity of 39.3 m/s. This produced an absolute error of 3.2 m/s and an accuracy of 91.14%. Additionally, the model successfully reconstructed a physically realistic flight path. However, the results display a higher variance in the trajectory data, as a predicted launch angle of 39.2 degrees is relatively steep for a standard free kick.

Comparing the two test cases, AeroStrike AI demonstrated improved velocity estimates in the Cristiano Ronaldo free kick test case, with a significant difference in terms of accuracy (19.24% increase). This difference highlights the sensitivity of monocular trajectory when accounting for factors such as camera perspective and calibration accuracy.

Limitations

Despite successfully modeling the three dimensional motion of soccer balls, several limitations remain. For one, the accuracy of the framework is largely dependent on the quality of the input footage, with factors such as motion blur, video compression and camera resolution reducing the reliability of automated ball detection.

The reliance on a monocular approach presents several challenges, most notably the inability to measure depth information directly from a single camera. To address this challenge, AeroStrike AI implements geometric estimation using the size of the soccer ball and camera perspective.

Consequently, inaccuracies in camera estimation and calibration may lead to errors in reconstructed trajectories.

Lastly, external factors such as wind and air turbulence may result in discrepancies between simulated and observed trajectories.

Conclusion

This paper presented AeroStrike AI, a computational physics framework that reconstructs the three dimensional flight trajectory of a soccer ball from a monocular video. By combining automated ball tracking, geometric reconstruction, numerical optimization and aerodynamic simulation, the system is able to accurately estimate the flight characteristics using standard video footage. The framework aims to demonstrate that meaningful and thorough analysis can be achieved without specialized infrastructure or cutting edge technology.

Experimental evaluation effectively demonstrated the accuracy with which AeroStrike AI was able to successfully reconstruct physically realistic trajectories, while producing quantitative estimates. Despite the differences found between the predictive model and reference videos, the results showcase the framework's capabilities in extracting useful aerodynamic information.

Overall, AeroStrike AI illustrates the feasibility and impact that performing three dimensional trajectory reconstruction has in providing an accessible foundation for future development in sports analytics.